

YOSEF D. STIFANOS

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EDUCATION

University of Illinois at Urbana-Champaign

B.S. in Information Sciences + Data Science & Cybersecurity

Urbana, Illinois

August 2022 – May 2026

- **GPA:** 3.88/4.00, 6x Dean's List, NSBE Chicago Scholarship Recipient
- **Relevant Coursework:** Data Science 1&2, Machine Learning, Risk & Cybersecurity, Fundamentals of Information Security, Privacy & Information Technology, Database Concepts and Applications, Cybersecurity Concepts & Practices

EXPERIENCE

Caterpillar Inc.

Cybersecurity Engineer

Irving, Texas

March 2026 – Present

- Cut manual reconciliation across compliance data formats, as measured by faster, standardized control catalog outputs, by building Python automation to transform data across Excel, JSON, CSV, and OSCAL.
- **Achieved 100%+ of project goals across all 7 deliverables**, as measured by an estimated **\$100k+ in annual value and 15 hours/week saved per design team, by co-developing a multi-agent AI system** — PDF requirement extraction, Azure AI Search-based catalog research, automated BoM assembly, and an engineer-facing chatbot — that generates draft Bills of Materials in minutes instead of hours.

Freddie Mac

Technology Analyst

McLean, Virginia

May 2025 – July 2025

- Participated in planning and executing 2 tabletop exercises simulating cyber incidents, system outages, and recovery scenarios to stress-test enterprise resilience and incident response readiness.
- Mapped recovery dependencies across hundreds of enterprise applications, identifying critical systems and post-incident prioritization gaps to support business continuity planning.
- Produced a Power BI report for the Cyber/Technology Resilience team, synthesizing attack path analysis and resilience gap findings into actionable executive deliverables.

The Grainger College of Engineering

Network & Infrastructure Engineering Student Consultant

Urbana, Illinois

April 2025 – March 2026

- Provided AV/IT support across 100+ Grainger Engineering classrooms, resolving ~100 tickets over two semesters to maintain high-availability instructional environments.
- Deployed and managed digital signage across 50+ displays college-wide using the Korbyt platform across multiple engineering buildings.
- Administered ~150 IP address allocations using IPAM with zero documentation errors, ensuring accurate network records and reliable enterprise connectivity.

Google

Cybersecurity Scholar

Remote, California

August 2025 – December 2025

- Selected for the Mentor Me Collective x Grow with Google Scholarship Program, completing the Google Cybersecurity Professional Certificate with hands-on labs in Splunk, Wireshark, and Linux-based threat detection and incident response.

Vivacity Tech PBC

Software Quality Engineer Intern

Remote, St. Paul, Minnesota

January 2024 – June 2024

University of Illinois Technology Services

Information Technology Operations Analyst

Urbana, Illinois

September 2022 – February 2025

PROJECTS

NIST Password Generator + Passphrase Tool

Independent Owner

Urbana, Illinois

July 2025

- Integrated the Have I Been Pwned API to detect compromised passwords in real time using k-anonymity, ensuring plaintext credentials are never exposed.
- Leveraged Claude AI to validate cryptographic randomness logic and support rigorous security testing, ensuring implementation correctness against NIST SP 800-63B standards.
- **Tools:** HTML, JavaScript, Web Crypto API, Have I Been Pwned API, Cybersecurity Standards

Predictive Modeling on Game Sales and Playtime

Group Lead

Urbana, Illinois

December 2024

- Developed regression and classification models to forecast game sales and playtime behavior.
- Engineered features, trained/tested models, and evaluated performance using R², AUC, and RMSE.

ACTIVITIES AND LEADERSHIP

NSBE Member – Coordinated STEM workshops for local schools, increasing outreach by 30%.

SIGPwny – Participated in weekly red/blue security workshops and CTF competitions focused on real-world hacking techniques.

SKILLS

Languages: Python, JavaScript, HTML, CSS, SQL

Tools: Git, Jira, Jupyter Notebooks, Korbyt, MySQL, MongoDB, Wireshark, IPAM, Splunk, Windows, Linux, MacOS, LLM API Integration, Prompt Engineering

Frameworks/Libraries: Pandas, NumPy, Matplotlib, Scikit-learn

Certifications: Google Cybersecurity Professional Certificate

Cybersecurity: CIA Triad, OSI Model, MITRE ATT&CK, ServiceNow IRM, Kali Linux, NMAP, SIEM, OSCAL, SBOM, Vulnerability Assessment, Patch Management, System Security Assurance